# Amirmohammad Mohammadi

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## EDUCATION

**Texas A&M University,** College Station, Texas — May 2027 (anticipated)
Doctor of Philosophy in Computer Engineering
Thesis: Distribution-guided Parameter-efficient Fine-tuning of Foundation Models for Waveform-based Sensing

**Sharif University of Technology,** Tehran, Iran — February 2021
Master of Science in Electrical Engineering
Thesis: Design and Evaluation of an Integrated Biomedical System for Human Stress Detection and Monitoring

**University of Tabriz,** Tabriz, Iran — September 2018
Bachelor of Science in Electrical Engineering
Final project: Development of a Data Logger for Sensors of Smartphones

## PROFESSIONAL EXPERIENCE

Texas A&M University, College Station, Texas — September 2022 – Present
***Graduate Research Assistant***

- **Built** a few-shot whole slide image classification pipeline with explainable AI capability to investigate parameter-efficient fine-tuning of vision-language models for alcoholic liver disease diagnosis.
- **Proposed** a neighborhood feature pooling layer that improved remote sensing image classification accuracy by up to **2.5 percentage points** over standard pooling with near-zero added overhead.
- **Reduced** the fine-tuning parameters of Transformer-based foundation models by over **10%** compared to conventional adapters by developing a distribution-aware adaptation algorithm.
- **Raised** the classification accuracy of a convolutional deep learning model by **7 percentage points** by designing a time-frequency feature engineering pipeline for audio applications.
- **Established** a cross-domain transfer learning benchmark showing that ImageNet-pretrained models outperform audio-pretrained models by **3 percentage points** in acoustic classification.
- **Reduced** ground truth data requirements by a factor of **15** for physiological time-series signals by developing physics-informed neural networks for cuffless blood pressure measurement.
- **Disseminated** research findings through publications, presentations, and reports to funding agencies.

Sharif University of Technology, Tehran, Iran — July 2019 – February 2021
***Graduate Student Researcher***

- **Designed** an ECG + EDA electrical circuit for a wearable sensor (BLE SoC) with a **3×** longer battery life compared to alternatives and delivered **94%** mental stress detection accuracy across 18 participants.

## SKILLS

- **Tools:** Python; PyTorch; scikit-learn; NumPy; Pandas; Matplotlib; MATLAB; C; Hugging Face; Git; Linux; VS Code; Jupyter Notebook; High-performance Computing.
- **Expertise:** Applied AI; Foundation Models; Deep Learning Architectures & Techniques; Generative AI; Physics-Informed Machine Learning; Multi-Modal Learning; AI Hardware; Efficient AI.

## AWARDS

Fully funded research assistantship, $152,000 January 2024 – August 2026

- Awarded by **Massachusetts Institute of Technology (MIT) Lincoln Laboratory** to advance parameter-efficient fine-tuning techniques of foundation models for waveform based sensing.

Travel grant, $500 May 2026

- Awarded by **IEEE Geoscience and Remote Sensing Society** to present at conference.

Travel grant, $1000 May 2026

- Awarded by **TAMU Electrical & Computer Engineering Department** to present at conference.

Fully funded research assistantship, $79,000 September 2022 – December 2023

- Awarded by **National Institutes of Health (NIH)** to investigate physics-informed neural networks for improved cuffless blood pressure estimation.

Tuition waiver for M.Sc in electrical engineering September 2018 – February 2021

- Awarded based on top-tier performance in the Iran Nationwide **University Entrance Exam**.

Tuition waiver for B.Sc. in electrical engineering September 2014 – August 2018

- Awarded based on top-tier performance in the Iran Nationwide **University Entrance Exam**.

## PEER-REVIEWED PUBLICATIONS

- Neighborhood Feature Pooling for Remote Sensing Image Classification., *Orvati Nia, F.,* ***Mohammadi, A.****, Al Kharsa, S., Naikare, P., Hampel-Aria, Z., & Peeples, J.*, (2026). Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. [link]
- Histogram-based Parameter-efficient Tuning for Passive Sonar Classification., ***Mohammadi, A.****, Carreiro, D., Van Dine, A., & Peeples, J.,* (2026). IEEE International Geoscience and Remote Sensing Symposium. [link]
- Investigation of Time-Frequency Feature Combinations with Histogram Layer Time Delay Neural Networks., ***Mohammadi, A****., Masabarakiza, I., Barnes, E., Carreiro, D., Van Dine, A., & Peeples, J.,* (2025). IEEE OCEANS. [link]
- Cross-Domain Knowledge Transfer for Underwater Acoustic Classification Using Pre-trained Models., ***Mohammadi, A****., Kelhe, T., Carreiro, D., Van Dine, A., & Peeples, J.,* (2025). IEEE OCEANS. [link]
- Physics-informed neural networks for modeling physiological time series for cuffless blood pressure estimation., *Sel, K.,* ***Mohammadi, A.****, Pettigrew, R. I., & Jafari, R.,* (2023). Nature NPJ Digital Medicine, 6(1), 110. [link]
- An integrated human stress detection sensor using supervised algorithms., ***Mohammadi, A.****, Fakharzadeh, M., & Baraeinejad, B.,* (2022). IEEE Sensors Journal, 22(8), 8216-8223. [link]

## MANUSCRIPTS UNDER REVIEW

- Structural and Statistical Audio Texture Knowledge Distillation for Acoustic Classification., *Ritu, J.,* ***Mohammadi, A.****, Carreiro, D., Van Dine, A., & Peeples, J.*, [link]
- Parameter-efficient Dual-encoder Architecture with Differentiable Choquet Integral Fusion for Underwater Acoustic Classification., ***Mohammadi, A.****, Van Dine, A., & Peeples, J.*,

## PRESENTATIONS

- Neighborhood Feature Pooling for Remote Sensing Image Classification., *Orvati Nia, F.,* ***Mohammadi, A.****, Al Kharsa, S., Naikare, P., Hampel-Aria, Z., & Peeples, J.*, (2026). IEEE/CVF Winter Conference on Applications of Computer Vision (2026). Tuscon, AZ. (poster & oral presentation)
- Investigation of Time-Frequency Feature Combinations with Histogram Layer Time Delay Neural Networks., ***Mohammadi, A****., Masabarakiza, I., Barnes, E., Carreiro, D., Van Dine, A., & Peeples, J.,* (2024). ECE Spring 2024 Poster Event. College Station, TX. (poster presentation)
- Physics-Informed Neural Networks for Modeling Cardiovascular Dynamic., ***Mohammadi, A.****, Sel, K., Pettigrew, R. I., & Jafari, R.* (2023). AI in Health Conference. Houston, TX. (poster presentation)

## EXTRACURRICULARS

- **Reviewer** for the 2023 IEEE International Conference on Acoustics, Speech and Signal Processing.
- **Reviewer** for the Expert Systems With Applications international journal.
- **Helper** for the 2024 IEEE International Conference on Acoustics, Speech and Signal Processing.
- **Evaluated** student performance and provided academic support for an electronics course.
- **Mentored** undergraduate and graduate students in designing & developing end-to-end AI projects.
- **Volunteered** for Night at Zach event to demonstrate lab projects and answer questions from attendees.
- **Delivered** 3 Machine Learning lectures (neural networks/backpropagation) to a class of 99 students.

## LANGUAGES

**English** (proficient); **Farsi** (native/bilingual); **Azerbaijani Turkish** (native/bilingual)